



HEALTH AND RETIREMENT STUDY
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DBS-based Cross-wave C-reactive Protein: 2006-2016

User Guide

Eileen Crimmins, University of Southern California
Jung Ki Kim, University of Southern California

Survey Research Center
Institute for Social Research
University of Michigan
Ann Arbor, Michigan

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Introduction

This documentation provides an update to previous documentation of C-Reactive Protein (CRP) from DBS blood-based biomarkers, specifically the NHANES equivalent values for all waves of DBS CRP from 2006 to 2016. In the previous data release, NHANES 2005-2006 and 2007-2008 were used to create NHANES equivalent values for HRS 2006 and 2008, NHANES 2009-2010 for HRS 2010, 2012, and 2014, and NHANES 2015-2016 for HRS 2016. However, we have since observed changes in NHANES 2009-2010 that appear to reflect assay modifications at NHANES rather than changes in the CRP value. Additionally, no CRP data are available for NHANES 2011-2012 and 2013-2014. In order to make HRS CRP NHANES equivalent values comparable over time, we have readjusted CRP and provide updated data, with NHANES equivalents now based on NHANES 2005-2008 for all HRS waves. We recommend use of these updated equivalent assay values for analysis of longitudinal change. It is important to note, however, that these values are all based on NHANES 2005-2008 data and so do not account for changes in the average levels of CRP in the population over time, but they will reflect individual changes in level over time.

Comparing Cross-wave Measures to Wave-specific CRP Values

Comparison of the original CRP values, NHANES 2005-2008 and the updated NHANES equivalent values for 2006, 2008, 2010, 2012, 2014, and 2016 are shown in Table 1 below along with the equations used to make these values. Figure 1 shows the distributions of the data before and after harmonization.

Table 1. Descriptive Statistics for CRP (weighted)

	N	Mean	SD	Min	Max
HRS					
2006 KCRP	5818	2.11	3.87	0.02	101.64
2008 LCRP	5777	2.32	5.07	0.01	112.14
2010 MCRP	7459	3.48	8.17	0.06	177.41
2012 NCRP	6783	4.23	10.20	0.06	244.67
2014 OCRP	6749	3.88	6.66	0.06	112.67
2016 PCRP	6869	3.79	5.47	0.09	20.00
NHANES					
2005-2008	4774	4.51	8.36	0.10	200.00
HRS NHANES Equivalent Value based on NHANES 2005-2008 (weighted)					
2006 KXCRP_ADJ	5818	4.45	8.95	0.03	255.12
2008 LXCRP_ADJ	5777	4.66	10.84	0.02	248.96
2010 MXCRP_ADJ	7459	4.65	11.58	0.07	259.08
2012 NXCRP_ADJ	6783	4.71	12.27	0.05	309.17
2014 OXCRP_ADJ	6749	4.38	8.04	0.05	143.04
2016 PXCRP_ADJ	6869	4.15	6.09	0.09	22.30

Conversion to NHANES Equivalent Values

KXCRP_ADJ = $\exp(0.6659644 + 1.055037 \cdot \log(\text{KCRP}))$;

LXCRP_ADJ = $\exp(0.6411116 + 1.033143 \cdot \log(\text{LCRP}))$;

MXCRP_ADJ = $\exp(0.2295684 + 1.028790 \cdot \log(\text{MCRP}))$;

NXCRP_ADJ = $\exp(0.0025181 + 1.042087 \cdot \log(\text{NCRP}))$;

OXCRP_ADJ = $\exp(0.0337862 + 1.043364 \cdot \log(\text{OCR}))$;

PXCRP_ADJ = $\exp(0.0571446 + 1.017251 \cdot \log(\text{PCRP}))$;

Figure 1. Distribution of CRP in Originally released data and in Harmonized Data

